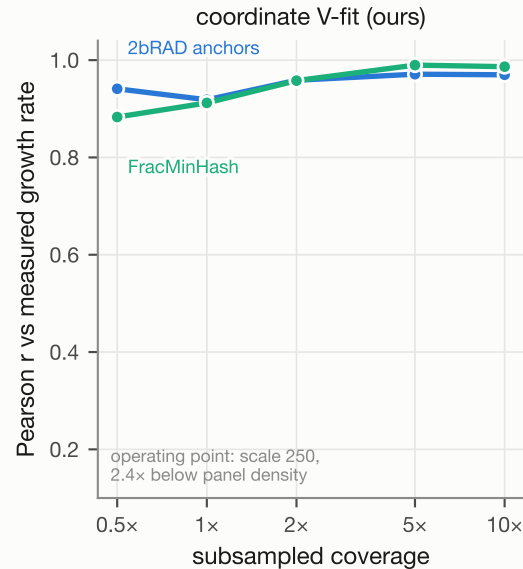
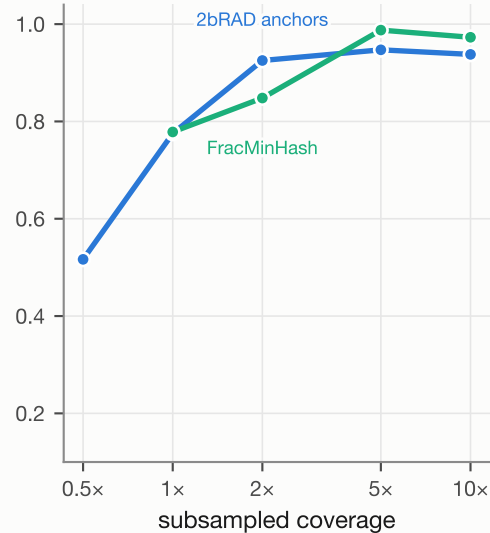
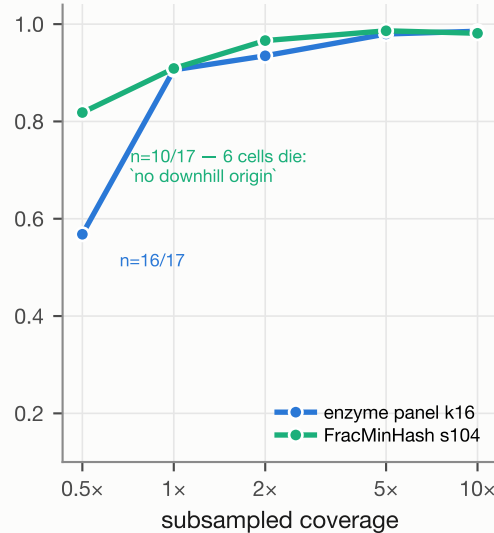
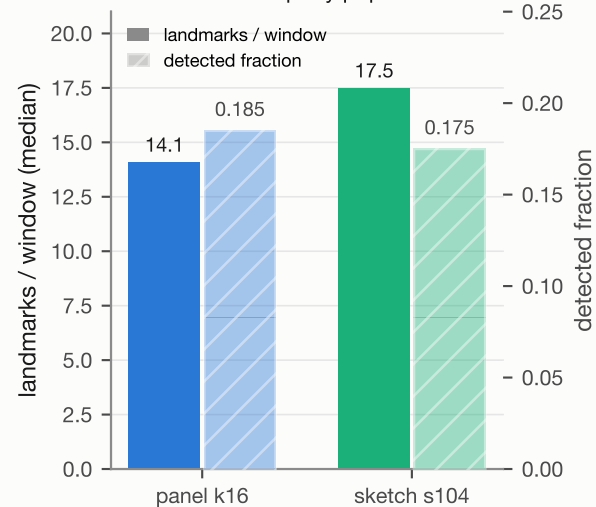


a



rank regression (iRep / Pilea)

b density-matched (s104 vs k16)
same estimator, same densityc mechanism at 0.5×:
windows equally populated

a, the full 2×2 at the Pilea operating point — scale 250, a landmark density 2.4× below the enzyme panel — which is where the sketch penalty lives; the interaction is a fact about this operating point, not a property of the sketch. b, the same comparison after density matching (FracMinHash scale 104 vs the 16-enzyme panel; 9,645 vs 9,422 landmarks/Mb): at $\geq 1\times$ the two landmark modes are tied at every depth, and at 0.5× the sketch scores higher on the cells it survives (r 0.82–0.84, n = 10–11, vs 0.50–0.57, n = 15–16) — but it loses 6 of 17 cells to `no downhill origin`. c, the mechanism at 0.5×: windows are populated almost identically in the two modes (median 17.5 vs 14.1 landmarks/window; detected fraction 0.175 vs 0.185) — deterministic anchors are not what keeps windows populated. The panel's low-coverage robustness comes from redundancy across its 16 strata: a single-strata sketch dies outright when its one stratum fits no gradient and no fusion can rescue it, while the panel still answers if any of its 16 strata do.

(panel a: at 1× the coordinate fit is worth +0.14 r on 2bRAD anchors but only +0.13 on a FracMinHash sketch; interaction +0.01.)